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RESEARCH ARTICLE

Statistical Forensics of the 2026 Brazilian Presidential Election at the Polling-Section Level: a CRISP-DM Application to the Electoral Court's Open Data

Forense Estatística da Eleição Presidencial Brasileira de 2026 por Seção Eleitoral: uma Aplicação da Metodologia CRISP-DM aos Dados Abertos do TSE

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Abstract: Brazil publishes the vote count of every polling section, which allows anyone to test whether the counts follow the statistical regularities expected from an unmanipulated tally. We applied eight forensic tests to the 2026 presidential election (first round on 4 October, 499,206 sections; runoff on 25 October, 499,205 sections), using the Superior Electoral Court (TSE) open data and the six phases of CRISP-DM. Every test was first validated on synthetic sections with and without inserted frauds. Votes matched turnout in every section. Last-digit, second-digit and integer-percentage tests stayed within the expected range for both finalists; sections at the same polling place were only 1.23–1.26 times more dispersed than random allocation of voters; and the first-round profile explained 93% of the within-municipality variance of the runoff ($R^2 = 0.93$). The turnout–vote fingerprint showed no smear toward the 100%/100% corner, and within-municipality turnout was *negatively* correlated with the winner's share ($r = -0.07$). Crossing the turnout change between rounds with the unexplained change in vote profile produced a small monotonic gradient (Spearman $\rho = 0.045$; at most 0.22 percentage points), with opposite signs in 9 of 28 units; among the 158 extreme sections that gained voters, 88 shifted toward one finalist and 70 toward the other (binomial $p = 0.18$). The remaining outliers (about 0.01% of sections) concentrate in indigenous villages, prisons, sections abroad and a few urban sections, and mostly repeat across rounds. We list the sections for manual check against the printed tally and propose twelve further tests, including a full recount of the digital vote record.
Keywords: Election forensics — CRISP-DM — Open government data — Electronic voting — Benford's law — Electoral integrity

Resumo: O Brasil publica a contagem de votos de cada seção eleitoral, o que permite testar se as contagens seguem as regularidades estatísticas esperadas em uma apuração sem manipulação. Aplicamos oito testes forenses à eleição presidencial de 2026 (1º turno em 4 de outubro, 499.206 seções; 2º turno em 25 de outubro, 499.205 seções), com os dados abertos do Tribunal Superior Eleitoral (TSE) e as seis fases do CRISP-DM. Cada teste foi validado antes com seções sintéticas com e sem fraudes inseridas. Os votos coincidiram com o comparecimento em todas as seções. Os testes de último dígito, de segundo dígito e de percentuais inteiros ficaram dentro do esperado para os dois finalistas; seções do mesmo local de votação foram apenas 1,23 a 1,26 vez mais dispersas que uma alocação aleatória de eleitores; e o perfil do 1º turno explicou 93% da variância do 2º turno dentro do município. A "impressão digital" comparecimento–voto não mostrou deslocamento para o canto 100%/100%, e o comparecimento dentro do município foi *negativamente* correlacionado com a votação do vencedor ($r = -0,07$). O cruzamento entre a variação de comparecimento e a mudança não explicada de perfil entre turnos mostrou um gradiente pequeno (Spearman $\rho = 0,045$; no máximo 0,22 ponto percentual), com sinal oposto em 9 de 28 unidades; entre as 158 seções extremas que ganharam eleitores, 88 se deslocaram para um finalista e 70 para o outro (binomial $p = 0,18$). Os casos extremos restantes (cerca de 0,01% das seções) concentram-se em aldeias indígenas, presídios, seções no exterior e poucas seções urbanas, e em geral se repetem nos dois turnos. Listamos as seções para conferência no Boletim de Urna e propomos doze novos testes, entre eles a recontagem integral do Registro Digital do Voto.
Palavras-Chave: Forense eleitoral — CRISP-DM — Dados abertos — Urna eletrônica — Lei de Benford — Integridade eleitoral

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1. Introduction

Brazil publishes the result of every polling section (*seção eleitoral*), so anyone can reproduce the national tally and test whether the counts behave as an unmanipulated count should. For the 2026 general election the Superior Electoral Court (TSE) released, besides the tally sheet (*Boletim de Urna*, BU), the digital vote record (*Registro Digital do Voto*, RDV) and the log of each voting machine, as well as consolidated files in its Open Data Portal [1].

Election forensics is the branch of statistics that searches such data for the marks that manipulation tends to leave: invented numbers, rounded percentages, coordinated inflation of turnout in some sections, or swings that only a few sections show [2, 3]. In Brazil, Figueiredo Filho, Silva and Carvalho [4] applied five such tests to the 2018 presidential election, and an election-fingerprint analysis of the 2022 runoff found no statistical sign of ballot stuffing or voter rigging [5].

This article applies and extends that line of work to the 2026 presidential election. Three research questions guide the study: **RQ1** — do section-level counts follow the regularities expected from an unmanipulated tally?; **RQ2** — where there are deviations, are they systematic or isolated, and do they have a plausible demographic explanation?; and **RQ3** — are the changes in turnout and in vote profile between the first round and the runoff consistent with each other?

Contributions (i) a reproducible CRISP-DM pipeline, in R, that validates every test on synthetic data with known inserted frauds before using it on real data; (ii) the use of *local* references — the other sections of the same polling place or municipality — instead of national medians, which removes the regional false alarms we found in a first version; (iii) a new cross-round test (T8) that relates the change in turnout of each section to the part of its vote change that the first-round composition does not explain, with a symmetry test between the two finalists; and (iv) a list of flagged sections, classified by the name of the polling place, ready for manual verification.

The study is statistical and exploratory. An unusual pattern indicates where to look; it does not prove irregularity. Likewise, the absence of a pattern does not prove that no irregularity occurred.

2. Theoretical Background

Each forensic test starts from a regularity that honest counts tend to follow and that counts invented or altered by people tend to violate. Table 1 summarizes the tests used here and the main known limitation of each.

The first-digit Benford test was not used. Sections have bounded size, roughly between 100 and 400 voters, and in that range the law does not hold even for clean data [9].

Two statistical cautions run through the whole study. First, with hundreds of thousands of sections any tiny deviation yields $p < 0.001$, so effect size matters more than the p-value. Second, when testing section by section, the false discovery rate must be controlled [16] and the over-dispersion between sections must be corrected; outliers are measured with robust z-scores based on the median and the median absolute deviation [17].

3. Related Work

The literature on election forensics can be organized into four strands: digit tests, distributional tests on vote shares and turnout, cross-round or cross-election comparisons, and supervised learning on synthetic frauds. Table 2 compares the studies closest to ours.

Digit tests Mebane [8] proposed the second-digit Benford test (2BL) for precinct counts in United States presidential elections. Pericchi and Torres [10] applied first- and second-digit tests, plus a version restricted to bounded units, to elections in the United States, Puerto Rico and Venezuela; the second-digit law was clearly rejected only in the 2004 Venezuelan recall referendum and only for electronic voting units. Beber and Scacco [6] moved the focus to last digits and digit pairs, exploiting the biases people show when inventing numbers. Deckert, Myagkov and Ordeshook [9] showed that Benford-type tests can fail in both directions, and an experiment by Mack and Stoetzer [7], in which participants were paid to alter tally sheets without being detected, found that last-digit tests do not reliably identify manipulated counts. These results are why we treat T2 and T3 as weak evidence and calibrate them by simulation.

Vote-share and turnout distributions Klimek et al. [3] introduced the turnout–vote *fingerprint* and a model of ballot stuffing and voter rigging; the same group later found fingerprints of ballot stuffing in about 11% of Turkish polling stations in the 2017 referendum [14]. Kobak, Shpilkin and Pshenichnikov [11] showed that falsified Russian results contain an excess of integer percentages, and Rozenas [12] proposed a resampled kernel-density method that separates such spikes from those created by small units. Leemann and Bochsler [18] argued for testing several empirical implications of different forms of fraud together and discussed how to handle the resulting multiple comparisons, a point we follow with the Benjamini–Hochberg procedure.

Cross-round comparisons Closest to our T6 and T8, Klimek, Aykaç and Thurner [13] compared the two rounds of the 2023 Turkish presidential election at the ballot-box level and found that districts with few ballot boxes were more likely to show large vote swings toward the incumbent between rounds. Their test is about the size of the swing in remote units; ours relates the swing that the first round does not explain to the change in turnout and tests whether extreme cases favour one side.

Machine learning on synthetic frauds Cantú and Saiegh [19] trained a classifier on synthetic clean and manipulated elections and applied it to Argentina in the 1930s; Zhang, Alvarez and Levin [20] used random forests trained on synthetic

Table 1. Forensic tests used in this study

Test	Expected regularity	Known limitation	Reference
T1 Zeros and 100%	Few unanimous sections where the candidate has intermediate support	Isolated communities are genuinely unanimous	[2]
T2 Last digit	Digits 0–9 with 10% each in counts with two or more digits	Biased when there are few votes per section	[6, 7]
T3 Second digit (2BL)	Benford distribution of the second digit, mean 4.187	Rejects under normal strategic voting	[8, 9, 10]
T4 Integer percentages	Frequency of integer percentages equal to binomial simulations	Simple fractions create spikes in small sections	[11, 12]
T5 Local homogeneity	Sections of the same polling place are similar	Voters are not randomly assigned to sections	This study (quasi-binomial)
T6 Cross-round consistency	Runoff share predicted by the section's own first round; stable turnout	Transit and transferred voters change the electorate	This study; [13]
T7 Fingerprint	No smear of turnout and leader share toward 100%/100%	Needs eligible voters per section	[3, 14]
T8 Turnout × profile	No link between extra turnout and unexplained shift	Ecological regression assumptions	This study; [15]

data to flag polling places in Argentina's 2015 election. Cantú [21] used convolutional neural networks on images of about 50,000 tally sheets from Mexico's 1988 election and found blatant alterations in about a third of them, showing the value of checking the physical record and not only the published numbers. Our validation step borrows the idea of synthetic frauds, but uses it to calibrate and debug classical tests rather than to train a classifier.

Brazil Figueiredo Filho, Silva and Carvalho [4] applied five tests (2BL, last-digit mean, frequency of last digits 0 and 5, correlation between vote share and turnout, and resampled kernel density) to the 2018 presidential election at the section and municipality levels and found no evidence of fraud. For the 2022 runoff, Klimek and Thurner [5] analysed 471,983 units and found no fingerprints of ballot stuffing or voter rigging, and Nogueira [22] discussed statistical methods and electoral integrity in the 2022 Brazilian elections. Other Brazilian applications of Benford's law target different objects: campaign donations [23] and Senate elections from 1994 to 2018 with a change of numerical base [24].

Gap Previous Brazilian studies analysed one round at a time, used national references, and did not report how each test behaves on data with known frauds. This study analyses both rounds, links them section by section, uses local references, validates every test on synthetic frauds before using real data, and identifies the flagged sections by the name of the polling place.

4. Methodology

The study followed the six phases of CRISP-DM [25, 26], with the evaluation phase carried out *before* the use of real data (Figure 1). All processing was done in R with the `data.table` package.

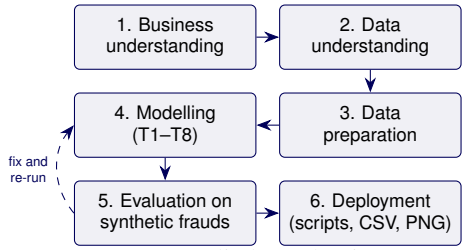


Figure 1. CRISP-DM as applied. Evaluation on synthetic data with known frauds came before the analysis of real data and fed corrections back into modelling.

Business understanding The question is whether section-level counts have regularities compatible with an unmanipulated tally. The success criterion is that each test separates clean data from data with inserted fraud, without false alarms.

Data understanding The sources are the files "President — vote by polling section" (`votacao_secao_2026_BR`) and "Count details by polling section" (`detalhe_votacao_secao_2026`), generated by the TSE on 26 October 2026 [1]. The first has one row per section, round and candidate, with state (UF), municipality, zone, section, polling place and votes; blank votes have code 95 and null votes code 96. The second has eligible voters and turnout per section. State files contain only state-level offices.

Data preparation Rows were filtered for the office of President and converted from Latin-1 to UTF-8. The TSE does not publish rows with zero votes, so each section was completed with zeros for all candidates of the round. The section key is UF + municipality + zone + section; valid votes exclude codes 95, 96 and 97. A load check compared both files: the sum of votes equals the reported turnout in *every* section of both rounds (125,275,835 votes in the first round and 128,236,074 in the runoff, for 158,745,502 eligible voters; turnout of 78.9% and 80.8%). The detail file lists 42 extra sections abroad in

Table 2. Related studies at the polling-unit level

Study	Election(s)	Methods	Main finding
Pericchi and Torres [10]	USA, Puerto Rico, Venezuela	First- and second-digit Benford, restricted version	2BL rejected only for Venezuela 2004 electronic units
Cantú and Saiegh [19]	Argentina 1931–1941	Naive Bayes trained on synthetic data	Historically irregular elections classified as fraudulent
Klimek et al. [3]	Several countries	Turnout–vote fingerprint, fraud model	Fingerprints of ballot stuffing in some countries
Beber and Scacco [6]	Several, incl. Nigeria	Last digits and digit pairs	Human-invented counts leave digit traces
Leemann and Bochsler [18]	Swiss referendum 2011	Several tests, multiple-comparison control	Irregularities in municipalities that destroyed ballots
Kobak et al. [11]	Russia	Integer percentages	Excess of round percentages
Klimek et al. [14]	Turkey 2017–2018	Fingerprint, fraud model	Ballot stuffing in about 11% of stations (2017)
Zhang et al. [20]	Argentina 2015	Random forest on synthetic data	Polling places at risk flagged
Cantú [21]	Mexico 1988	CNN on tally-sheet images	Alterations in about a third of tallies
Figueiredo Filho et al. [4]	Brazil 2018	Five tests (digits, turnout, kernel)	No evidence of fraud
Klimek and Thurner [5]	Brazil 2022, runoff	Fingerprint, ballot stuffing and rigging	No signs of fraud
Klimek et al. [13]	Turkey 2023, two rounds	Fingerprint, cross-round swings	Extreme swings in remote areas
This study	**Brazil 2026, two rounds**	**T1–T8 with synthetic validation, local references, cross-round turnout × profile**	**See Sections 5–6**

the first round and 43 in the runoff, all with eligible voters but zero turnout.

Modelling Eight tests, T1 to T8, described in Sections 5 and 6. References are local: the other sections of the same polling place or municipality, not the national median. Section-level tests use the Benjamini–Hochberg correction ($q < 0.001$) and robust z-scores [17].

Evaluation We generated 3,559 synthetic sections in the TSE layout, in two versions. One is clean. The other has three inserted frauds: 8 sections with 100% for one candidate, 300 sections with a forced integer percentage and 5 sections with inflated turnout. This step revealed three faults that were fixed before the real analysis: the theoretical digit references raised false alarms; the homogeneity test without dispersion adjustment flagged 75 clean sections; and turnout compared with the national median confused regional variation with anomaly. Table 3 shows the corrected tests.

Deployment Three reproducible scripts: download with integrity check, section-level distribution and the test battery, plus the T8 script. Results are written as CSV and PNG files, ready for checking against the tally sheets.

5. Results

Both finalists passed every aggregate regularity test. Deviations appear in few sections and, in general, repeat the same pattern in both rounds. We analysed 499,206 sections in the first round and 499,205 in the runoff.

Table 3. Validation on synthetic data (3,559 sections)

Test	Clean data	Inserted frauds
T1 Zeros and 100%	no alert	8 of 8 sections detected
T4 Integer %	ratio 0.95 ($p = 0.72$)	ratio 2.86 ($p < 0.01$)
T5 Local homogeneity	0 sections	8 of 8 detected
T6 Cross-round	0 sections	7 of 8 detected
T6 Turnout	0 sections	4 flagged, 3 inserted
T8 Turnout × profile	1 section (lost votes)	4 of 5 inflated, all "more votes, toward beneficiary"

5.1 Sections with 0% and 100% (T1)

There are more unanimous sections than the binomial model predicts, but the excess is small in absolute terms (Table 4).

Table 4. Unanimous sections (T1)

Round	Event	Obs.	Exp.	p (Poisson)
2nd	100% Lula	106	49.7	2.7×10^{-12}
2nd	100% F. Bolsonaro	4	2.3	0.20
1st	100% Lula	38	18.5	4.5×10^{-5}
1st	0 votes F. Bolsonaro	81	37.4	4.6×10^{-10}

Three facts point to communities with homogeneous vote rather than to manipulation. The sections lie mostly in the states of MG (26 sections), AM (23), MA (22), PA (21) and RS (11) and typically have between 10 and 150 valid votes

(median 81). The same sections are unanimous in both rounds (for instance AM 2690/280, MA 8958/170 and 179, and PA 4111/516 and 519), and some sections are unanimous for the other finalist in both rounds (RS 85839/81, PA 4200/44). The expected value underestimates unanimity because it assumes voters similar to those of neighbouring sections, which does not hold for villages or isolated communities.

5.2 Digit tests (T2 and T3)

The last digit is uniform for both finalists in both rounds (Figure 2). The theoretical Benford reference for the second digit always rejects ($p \approx 0$), but the simulated reference shows deviations of at most 0.12 percentage points (Table 5).

Table 5. Digit tests with simulated references

Rd.	Candidate	Last, p	2nd, p	2nd, max. dev. (p.p.)
1st	Lula	0.48	0.61	0.06
1st	F. Bolsonaro	0.67	0.015	0.09
2nd	Lula	0.76	0.11	0.10
2nd	F. Bolsonaro	0.23	0.0027	0.12

With eight simultaneous tests the Bonferroni threshold is 0.006. Only the second digit of Flávio Bolsonaro in the runoff falls below it, with a practically irrelevant effect. Mebane [8] already notes that 2BL reacts to ordinary strategic behaviour of voters.

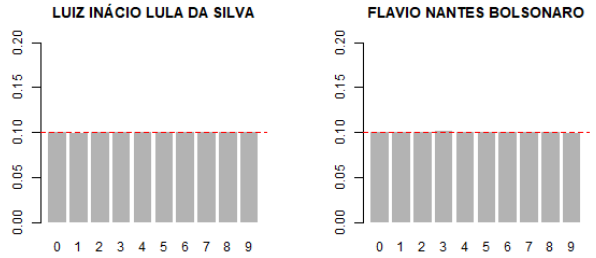


Figure 2. Last digit of the runoff counts for Lula (left) and Flávio Bolsonaro (right); the dashed line is the 10% expected under uniformity.

5.3 Integer percentages (T4)

There is no excess of integer percentages in the runoff: observed/expected ratio 0.995 for both finalists ($p = 0.72$ and 0.69). In the first round the ratios were 1.014 for Lula ($p = 0.059$) and 1.009 for Flávio Bolsonaro ($p = 0.18$). Among candidates with enough votes, only Augusto Cury reached the smallest p possible with 100 simulations (0.0099); his excess is about 250 sections, far below the Russian anomalies described by Kobak et al. [11]. The small peaks near 25%, 33% and 50% (Figure 3) come from simple fractions in small sections and are already included in the simulation.

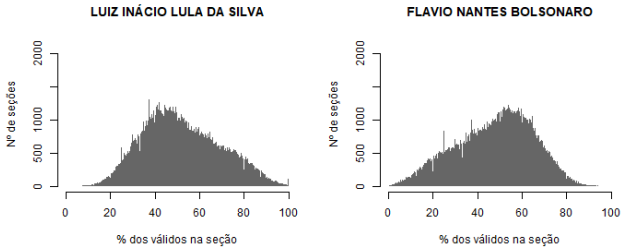


Figure 3. Distribution of the share of valid votes per section in the runoff (x axis: % of valid votes; y axis: number of sections).

5.4 Homogeneity within the polling place (T5)

Sections of the same building are almost as similar as they would be with randomly assigned voters: the dispersion ratio was 1.23–1.24 in the first round and 1.26 in the runoff for the finalists. After the quasi-binomial adjustment and the BH correction, between 67 and 85 sections per candidate and round differ by at least 20 percentage points, about 0.014% of the total. Almost all differ in the same direction and by a similar amount in both rounds (e.g. SP 62499/402, MG 41238/336, MA 9610/468), a pattern typical of a different electorate in the section, such as a special section or transferred voters.

5.5 Consistency between rounds (T6)

The first round of each section explains 93% of the variation of the runoff within the municipality ($R^2 = 0.9318$), and the scatter is a narrow band without clusters off the curve (Figure 4).

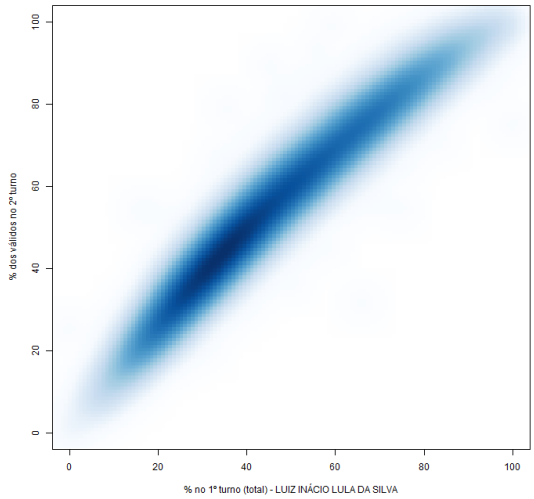


Figure 4. Lula's share in the first round (x axis, % of total) versus his share of valid votes in the runoff (y axis), per section; density plot.

Fourteen sections deviate from the prediction with robust $|z| \geq 6$ (Table 6). Six of them are in indigenous villages or indigenous lands, according to the polling-place name, and

one is abroad (Porto, Portugal). Four are in the same polling place in São Paulo (municipality 71072, zone 2), where Flávio Bolsonaro did better than predicted and the sections received 16–25% more votes in the runoff; they may be sections for voters in transit, since a voter can request transit voting in only one round. This hypothesis was not verified.

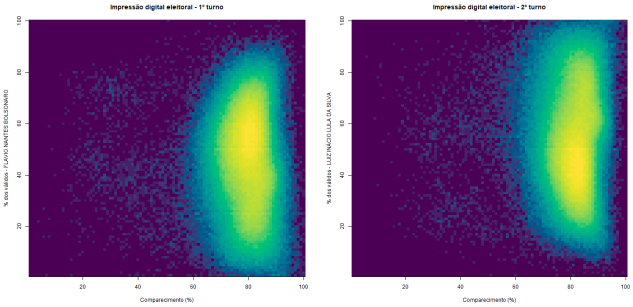
Table 6. Sections flagged by T6 ($|z| \geq 6$)

UF/mun./zone/sec.	Valid	Lula %	z	Polling place (type)
AC 1554/3/160	152	31.6	−14.7	Health post, Buaçú village (indig.)
AM 2097/18/37	155	54.2	13.1	Municipal school
AC 1031/3/95	221	78.7	12.6	Health unit, Maronawa village (indig.)
AM 2399/45/63	169	98.8	11.4	School, Piau village (indig.)
AC 1139/7/67	66	78.8	10.6	Nova Floresta village (indig.)
PI 11878/28/99	268	54.5	−9.7	Rural school
SP 71072/2/519	293	55.3	−7.1	School (same site)
SP 71072/2/539	312	64.1	−7.1	School (same site)
MA 8974/70/403	127	88.2	6.9	Zemuehan Tentenhar school (indig.)
BA 34134/170/262	60	66.7	6.8	School
SP 71072/2/491	308	51.3	−6.6	School (same site)
SP 71072/2/532	313	56.2	−6.5	School (same site)
PA 4723/80/254	162	97.5	6.3	Trincheira Bacajá school (indig.)
ZZ 30341/1/3576	251	64.9	6.0	Porto, Portugal (abroad)

Turnout change The first version of the turnout test compared each section with the national median and flagged 2,088 sections, which were in fact whole states (AC, AP, RR, MA and PA) where turnout fell in the runoff. The corrected version compares each section with its own municipality. It flagged 730 sections (0.15%) with robust $|z| \geq 6$ on the log-ratio of votes between rounds: 574 with *fewer* votes in the runoff and 156 with more. A keyword classification of the polling-place names (approximate; script `classificar_locais_t6.R`) found 54 prisons or juvenile detention units, 53 of them losing voters, consistent with releases and transfers of inmates between rounds; 53 indigenous villages, 40 of them gaining voters; 18 hospitals or care homes; and 18 sections abroad, 17 of them gaining voters. The flagged sections concentrate in PA (128), AC (85), AM (68), MA (67) and AP (64). The largest increases occurred in indigenous schools in PA and MS (e.g. PA 4111/18/519, from 21 to 35 votes; PA 5835/106/330, from 128 to 212) and in the embassy in Canberra (from 72 to 111).

5.6 Turnout–vote fingerprint (T7)

The two-dimensional histogram of turnout and leader share (Figure 5) shows a single compact cloud centred around 80% turnout, elongated along the vote axis, without the smear toward the 100%/100% corner that characterizes ballot stuffing [3, 14]. Within municipalities, turnout was *negatively* correlated with Lula's share ($r = -0.095$ in the first round and $r = -0.069$ in the runoff) and positively with Flávio Bolsonaro's ($r = 0.086$ and 0.069). Ballot stuffing in favour of the winner would produce the opposite sign. Sections with turnout $\geq 95\%$ and share $\geq 95\%$ for the same candidate are rare: 26 (first round) and 64 (runoff) for Lula and 1 and 2 for Flávio Bolsonaro, i.e. at most 0.013% of sections; they are likely the same small homogeneous sections found by T1, which remains to be confirmed. Integer turnout percentages were not in excess: 7,635 observed versus 7,581.5 expected in the first round ($p = 0.27$) and 7,088 versus 7,155.4 in the runoff ($p = 0.85$).



(a) First round, F. Bolsonaro **(b)** Runoff, Lula

Figure 5. Election fingerprints: turnout (x axis, %) versus the share of valid votes of the round leader (y axis, %); brighter means more sections.

6. Cross-analysis: Turnout × Change of Vote Profile (T8)

T8 tests whether sections whose turnout changed more than their neighbours' also changed their vote disproportionately. Inflating a section with votes for a single candidate leaves exactly that mark: more votes and a profile shifted toward the same side [3].

For each section present in both rounds with at least 30 votes in each, two measures are computed, both centred on the municipality median, using only municipalities with three or more sections:

$$\Delta C_s = \ln\left(\frac{V_{2,s}}{V_{1,s}}\right), \qquad \Delta P_s = 100\left(p_{2,s} - \hat{p}_{2,s}\right), \qquad (1)$$

where V is the section total in each round (including blank and null votes), p_2 is the share of finalist A (Lula) among the two finalists in the runoff, and $\hat{p}_2$ is the prediction of that share from the first-round composition of the same section (votes for A, for B, for other candidates, and blank or null), obtained from a transfer regression estimated by state [15]. ΔP thus measures the part of the profile change that the first round does not explain.

Deciles Figure 6 shows the mean of ΔP per decile of ΔC. The curve is not flat: it rises from −0.09 to +0.22 percentage points, so sections whose turnout grew most relative to their municipality shifted slightly toward Lula, and those whose turnout fell most shifted slightly toward Flávio Bolsonaro. The effect is statistically clear (non-overlapping 95% intervals at the extremes) but small: two tenths of a percentage point in the top decile, where turnout grew 5.1% above the municipal median.

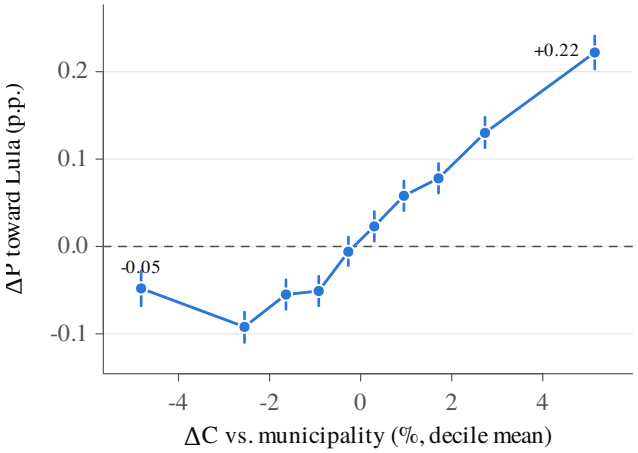


Figure 6. Mean unexplained shift toward Lula (ΔP, with 95% CI) per decile of turnout change relative to the municipality (ΔC); 498,952 sections.

Correlation by state The national Spearman correlation between ΔC and ΔP is $\rho = 0.045$ (0.031 without centring). It varies in sign across units (Figure 7): it is positive in 19 and negative in 9 of the 28 units, ranging from -0.10 (AP and AC) to $+0.16$ (sections abroad), with the largest positive values in the South and Centre-West (SC, MS, PR, DF, MT). A coordinated manipulation in favour of one candidate would push all states in the same direction; the mixed signs are more consistent with regional differences in which groups of voters were mobilized or demobilized between rounds.

Vote transfer The national ecological regression estimates that first-round voters of each finalist stayed with him (coefficients 1.03 and 1.01), voters of other candidates split evenly (0.44 to each), blank and null voters went mostly to Lula (0.63), and the extra runoff voters split 0.55 to Lula and 0.41 to Flávio Bolsonaro. In 9 units at least one coefficient fell outside $[0, 1]$ by more than 0.05, mostly Lula retention rates of 1.05–1.08 in SP, the South and the Centre-West. This indicates that the constant-transfer assumption of Goodman's regression does not fit there, a known limitation of ecological inference, not an impossible count.

Flagged sections and symmetry 594 sections had $|z| \geq 4$ in ΔC and $|z| \geq 2$ in ΔP (Figure 8). The pattern of ballot stuffing is "more votes, shifted toward the beneficiary". Among the 158 sections that gained voters, 88 shifted toward Lula and 70 toward Flávio Bolsonaro; the binomial test does not reject symmetry ($p = 0.18$). Among the 436 that lost voters, 263 shifted toward Lula and 173 toward Flávio Bolsonaro ($p = 1.9 \times 10^{-5}$); this asymmetry cannot come from added ballots and is consistent with the prisons, villages and urban sections found by T6. The four São Paulo sections of T6 appear again, within a group of ten sections of zone 2 of the same municipality (five at the same school and five at another polling place) that all gained votes and shifted toward Flávio Bolsonaro. A shared shift across two polling places reinforces

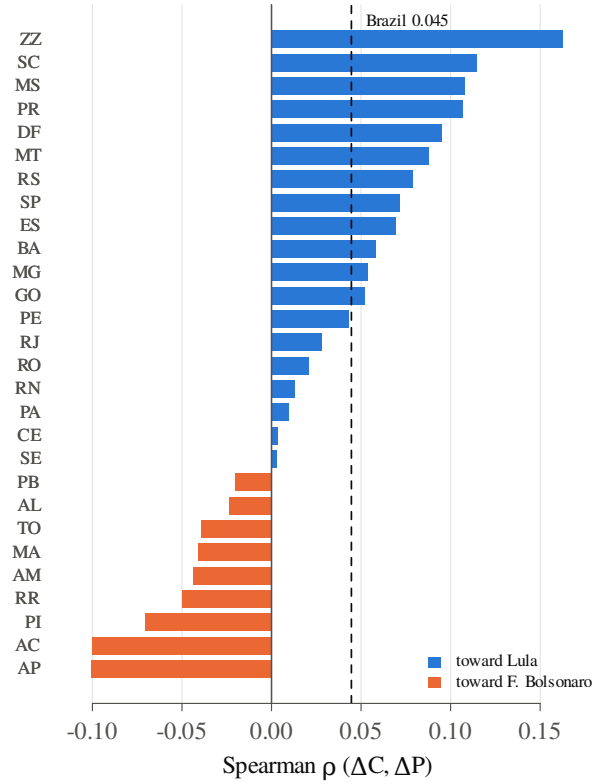


Figure 7. Spearman correlation between ΔC and ΔP by state (ZZ = abroad). Dashed line: national value.

the hypothesis of a change in the electorate (transit voters) rather than an anomaly specific to one candidate.

In the validation, T8 found 4 of the 5 inflated sections, and even with the inserted frauds the overall correlation stayed at -0.005: a few inflated sections do not move the correlation. The list of sections and the symmetry test are therefore the decisive outputs, and both point to no one-sided pattern.

7. Proposed Further Tests

The strongest tests stop looking only at votes and use the voting machine's own files and the profile of the electorate (Table 7). The first three answer directly whether the published result is what the machine recorded; the others refine the search for unusual patterns.

Tests 1 and 2 have an advantage that no statistical test has: if the RDV recount matches the BU in every section and all signatures are valid, it is demonstrated that the published result is the one recorded by the machines, without depending on any model or assumption about voters.

8. Discussion

Regarding **RQ1**, the 2026 section-level counts follow the expected regularities: digits, integer percentages, local homogeneity and the fingerprint behave as in clean data, and the runoff is predicted by the first round with $R^2 = 0.93$. Re-

Table 7. Proposed tests

#	Test	What it checks	TSE data	Method
1	BU × RDV recount	Tally sheet matches the votes recorded one by one	Machine files: BU and RDV	Decode with the published ASN.1 specification [27] and recount all sections; 0 divergences expected
2	Digital signature	Published files were not altered after the machine	Machine files: signatures and hashes	Verify the signature of every BU and RDV
3	Printed BU × published BU	Paper posted at the section equals the tallied file	BU QR code collected by observers	Stratified random sample by state, with CI for the divergence rate
4	Voting rhythm in the log	Votes recorded at a human pace, 8 a.m.–5 p.m.	Machine log	Time between votes, votes per hour, bursts, votes after closing; robust z
5	Releases without biometrics	Excess of voters enabled by code, without fingerprint	Machine log	Rate per section versus municipality, crossed with T8
6	Machine model	Vote does not depend on equipment model	Log/BU (UE2009 to UE2022)	Regression with municipality fixed effect; model effect should be zero
7	Electorate profile	Section vote consistent with age, sex and schooling	Electorate profile by section	Hierarchical beta-binomial model; posterior predictive checks
8	Panel 2022 → 2026	Abrupt changes in the same section between elections	2022 vote by section	Residual of 2022→2026 change, centred on municipality
9	Percentage spikes (RKD)	Excess of round percentages independent of section size	Vote by section	Resampled kernel density [12]
10	Spatial autocorrelation	Clusters of anomalous nearby sections	Polling-place coordinates	Local Moran (LISA) on T6 and T8 residuals
11	Machine-learning detection	Sections similar to simulated fraud scenarios	All of the above	Random forest on synthetic data [20, 19]
12	Local multimodality	Mixture of two vote regimes within a municipality	Vote by section	Hartigan dip test [28] per municipality, BH correction

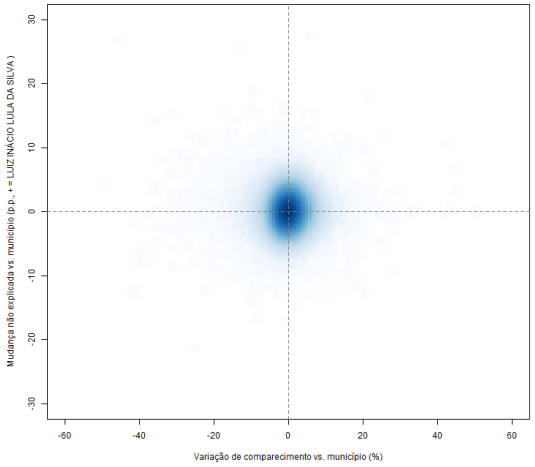


Figure 8. Unexplained shift toward Lula (y axis, p.p.) versus turnout change relative to the municipality (x axis, %); density plot.

garding **RQ2**, deviations are rare (about 0.01% of sections for vote tests and 0.15% for turnout change), most repeat in both rounds, and their polling-place names point to indigenous villages, prisons, care homes, sections abroad and a few urban sites, where the electorate is either homogeneous or changes between rounds. Regarding **RQ3**, turnout change and profile change are linked by a small, monotonic gradient whose sign varies across states, and extreme cases that gained voters are symmetric between the finalists.

These results agree with the earlier Brazilian studies for 2018 [4] and 2022 [5]. They differ from the Turkish case [13], where extreme cross-round swings in remote areas favoured the incumbent: in Brazil 2026 the remote sections with extreme changes do not favour one finalist systematically.

Threats to validity *Construct validity*: the tests detect specific forms of manipulation — invented numbers, rounding and inflated sections. A small, spread-out alteration that preserves all these regularities would not be seen; tests 1 to 3 of Section 7 address this. *Internal validity*: binomial references assume similar voters within a polling place, which underestimates unanimity in isolated communities (T1); the ecological regression of T8 does not fit in 9 units; and the classification of polling places by keywords is approximate. *External validity*: the results refer to the data published by the TSE on 26 October 2026; later updates may change marginal counts. *Reliability*: all steps are scripted and the outputs are saved as files, so the analysis can be reproduced by third parties.

9. Conclusion and Future Work

The section-level data of the 2026 Brazilian presidential election do not show the typical patterns of count manipulation. Digit and integer-percentage tests are within the expected range for both finalists; sections of the same polling place are homogeneous; the runoff of each section is predicted by its

first round; the fingerprint has no smear toward full turnout and full support; and the sections that gained voters between rounds do not favour one finalist. The deviations found are rare, mostly repeated in both rounds, and located in sections whose electorate is homogeneous or changes between rounds.

We recommend: (i) checking the 14 sections flagged by T6 against the tally sheets in the TSE "Machine Files" portal, and verifying whether the four São Paulo sections are transit-voting sections; (ii) confirming the type of the flagged polling places with the official register rather than keywords; (iii) prioritizing the full BU $\times$ RDV recount and signature verification, the only tests that confirm the result directly; (iv) publishing code, data version and results openly; and (v) communicating the findings as statistical: a flagged section is a point to check, not evidence of fraud.

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Author contributions

The author designed the study, wrote the analysis scripts, ran the analyses and wrote the manuscript.

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